

CMP-08 - Knowledge Service

Maintain provenance-aware knowledge representations and retrieval while distinguishing claims, evidence, confidence, validity, and authoritative state.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 10. Authoritative state: Knowledge Graph, knowledge assertions, provenance links, validity and confidence metadata, retrieval indexes. Explicitly non-owned: Evidence Objects, Experience Ledger, Memory Graph, authoritative operational state.

Critical invariant

No assertion becomes Knowledge without provenance, domain, time, confidence, and challenge semantics.

Logical interfaces

CMP-08-IF-01 / Command / Admit Knowledge Assertion

Command `Admit Knowledge Assertion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-02 / Command / Revise Knowledge Link

Command `Revise Knowledge Link` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-03 / Command / Invalidate Assertion

Command `Invalidate Assertion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-04 / Command / Rebuild Index

Command `Rebuild Index` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-05 / Query / Query Knowledge

Query `Query Knowledge` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-06 / Query / Trace Provenance

Query `Trace Provenance` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-07 / Query / Get Validity Context

Query `Get Validity Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-08 / Event / Knowledge Admitted

Event `Knowledge Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-09 / Event / Assertion Invalidated

Event `Assertion Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-08-IF-10 / Event / Index Rebuilt

Event `Index Rebuilt` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Candidate > Admitted > Qualified > Challenged > Superseded or Retired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: provenance loss, stale assertion, index-authority confusion, causation inference, unbounded disclosure. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-08-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-08-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-08-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-08-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-08-TST-005 - Critical invariant. No assertion becomes Knowledge without provenance, domain, time, confidence, and challenge semantics. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-08-TST-006 - Failure containment. Inject provenance loss, stale assertion, index-authority confusion, causation inference, unbounded disclosure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-08-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-08-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-08-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-08-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.